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ROLL.NO.: 251256
DEPT: B.SC.(H) COMPUTER SCIENCE
DATA MINING ASSIGNMENT 1

```
DM.py > ... D:\DM assignment 1\DM.py
1 import pandas as pd
2 import random
3
4 names=["Aarav","Vivaan","Aditya","Vihaan","Arjun","Reyansh","Muhammad","Sai","Ayaan","Krishna",
5        "Ishaan","Shaurya","Atharv","Rudra","Ansh","Advik","Dhruv","Kabir","Aryan","Aarush","Raghav","Shivansh","Om",
6        "Pranav","Yuvraj","Aarav","Vivaan","Aditya"]
7
8 surnames=["Sharma","Verma","Gupta","Mehta","Jain","Kapoor","Malhotra","Chopra","Bhatia","Agarwal",
9           "Singh","Kumar","Reddy","Patel","Nair","Iyer","Rao","Chatterjee","Mukherjee","Das","Bose","Ghosh","Sen","Roy","Dutta",
10          "Saxena","Choudhary","Trivedi","Joshi","Shah"]
11
12 data = []
13
14 for i in range(1,201):
15     row={
16         "ID":i,
17         "Name":random.choice(names),
18         "Surname":random.choice(surnames),
19         "Maths":random.randint(0,50),
20         "Science":random.randint(0,50),
21         "English":random.randint(0,50),
22         "History":random.randint(0,50),
23         "Geography":random.randint(0,50)
24     }
25     data.append(row)
26
27 dataframe=pd.DataFrame(data)
28
29 dataframe.to_csv("student_data.csv",index=False)
30 print("CSV file 'student_data.csv' has been created successfully.")
31
```

```
30 print("CSV file 'student_data.csv' has been created successfully.")
31
32 # (a) Average marks of all subjects
33 print("\nAverage marks of all subjects:")
34 print(dataframe[["Maths","Science","English","History","Geography"]].mean())
35
36 # (b) overall Topper
37 dataframe["Total"] = dataframe[["Maths","Science","English","History","Geography"]].sum(axis=1)
38 topper = dataframe.loc[dataframe["Total"].idxmax()]
39 print("\nOverall Topper:")
40 print(topper[["ID","Name","Surname","Total"]])
41
42 # (c) Subject-wise Toppers
43 print("\nSubject-wise Toppers:")
44 subjects = ["Maths","Science","English","History","Geography"]
45
46 for subject in subjects:
47     topper = dataframe.loc[dataframe[subject].idxmax()]
48     print(f"{subject} Topper: ID: {topper['ID']}, Name: {topper['Name']}, Surname: {topper['Surname']}, Marks: {topper[subject]}")
49
50 # (d) Students scoring less than 15 in any subject
51 low_marks=dataframe[(dataframe["Maths"] < 15) | (dataframe["Science"] < 15) | (dataframe["English"] < 15) | (dataframe["History"] < 15) | (dataframe["Geography"] < 15)]
52 print("\nStudents scoring less than 15 in any subject:")
53 print(low_marks[["ID","Name","Surname"]])
54
55 low_marks.to_csv("low_marks_students.csv",index=False)
56
```

Output:

```
(.venv) PS D:\DM assignment 1> python DM.py
CSV file 'student_data.csv' has been created successfully.

Average marks of all subjects:
Maths      23.240
Science    23.175
English     23.765
History     25.830
Geography   25.925
dtype: float64

Overall Topper:
ID          78
Name        Vivaan
Surname      Choudhary
Total       214
Name: 77, dtype: object

Subject-wise Toppers:
Maths Topper: ID: 144, Name: Shaurya, Surname: Trivedi, Marks: 50
Science Topper: ID: 25, Name: Shivansh, Surname: Nair, Marks: 50
English Topper: ID: 70, Name: Vivaan, Surname: Gupta, Marks: 50
History Topper: ID: 118, Name: Aarav, Surname: Kumar, Marks: 50
Geography Topper: ID: 13, Name: Aarav, Surname: Ghosh, Marks: 50

Students scoring less than 15 in any subject:
   ID  Name  Surname
0    1  Pranav    Nair
1    2   Dhruv     Das
2    3  Aditya Chatterjee
3    4   Ayaan   Sharma
4    5  Raghav   Bhatia
..  ...   ...      ...
195 196  Vivaan   Chopra
196 197  Reyansh   Patel
197 198     Om   Agarwal
198 199   Rudra Choudhary
199 200  Aditya Chatterjee

[168 rows x 3 columns]
(.venv) PS D:\DM assignment 1> |
```

Student.csv File

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A1

ID

	A	B	C	D	E	F	G	H	I	J	K	
1	ID	Name	Surname	Maths	Science	English	History	Geography				
2	1	Pranav	Nair	20	23	27	26	6				
3	2	Dhruv	Das	1	11	28	18	42				
4	3	Aditya	Chatterjee	7	15	3	6	39				
5	4	Ayaan	Sharma	42	11	11	37	18				
6	5	Raghav	Bhatia	15	12	45	42	15				
7	6	Anay	Dutta	35	36	8	39	42				
8	7	Shaurya	Ghosh	37	14	29	28	20				
9	8	Vivaan	Iyer	12	33	27	39	29				
10	9	Shaurya	Choudhary	6	26	40	21	45				
11	10	Vivaan	Iyer	35	18	30	23	39				
12	11	Dhruv	Jain	41	42	42	22	0				
13	12	Yuvraj	Patel	0	23	22	17	46				
14	13	Aarav	Ghosh	0	12	5	43	50				
15	14	Aditya	Nair	13	8	5	19	30				
16	15	Sai	Kumar	0	44	5	31	42				
17	16	Shivansh	Mehta	0	5	20	1	15				
18	17	Ayaan	Singh	12	11	24	48	30				
19	18	Shivansh	Saxena	43	12	48	29	12				
20	19	Dhruv	Malhotra	6	14	23	22	38				
21	20	Ansh	Mukherjee	12	0	23	22	46				
22	21	Raghav	Bose	15	24	20	17	16				
23	22	Muhamma	Chopra	35	42	49	2	14				
24	23	Sai	Trivedi	47	24	16	28	11				
25	24	Reyansh	Bose	45	30	1	3	36				
26	25	Shivansh	Nair	24	50	10	34	47				
27	26	Pranav	Roy	37	9	39	46	45				
28	27	Aditya	Chatterjee	13	25	28	17	28				

<

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student_data

Sheet1

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Students marks less than 15 :

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Clipboard Font Alignment Number												
A1 ID												
	A	B	C	D	E	F	G	H	I	J	K	L
	ID	Name	Surname	Maths	Science	English	History	Geography	Total			
1	1	Pranav	Nair	20	23	27	26	6	102			
2	2	Dhruv	Das	1	11	28	18	42	100			
3	3	Aditya	Chatterjee	7	15	3	6	39	70			
4	4	Ayaan	Sharma	42	11	11	37	18	119			
5	5	Raghav	Bhatia	15	12	45	42	15	129			
6	6	Anay	Dutta	35	36	8	39	42	160			
7	7	Shaurya	Ghosh	37	14	29	28	20	128			
8	8	Vivaan	Iyer	12	33	27	39	29	140			
9	9	Shaurya	Choudhary	6	26	40	21	45	138			
10	11	Dhruv	Jain	41	42	42	22	0	147			
11	12	Yuvraj	Patel	0	23	22	17	46	108			
12	13	Aarav	Ghosh	0	12	5	43	50	110			
13	14	Aditya	Nair	13	8	5	19	30	75			
14	15	Sai	Kumar	0	44	5	31	42	122			
15	16	Shivansh	Mehta	0	5	20	1	15	41			
16	17	Ayaan	Singh	12	11	24	48	30	125			
17	18	Shivansh	Saxena	43	12	48	29	12	144			
18	19	Dhruv	Malhotra	6	14	23	22	38	103			
19	20	Ansh	Mukherjee	12	0	23	22	46	103			
20	22	Muhamma	Chopra	35	42	49	2	14	142			
21	23	Sai	Trivedi	47	24	16	28	11	126			
22	24	Reyansh	Bose	45	30	1	3	36	115			
23	25	Shivansh	Nair	24	50	10	34	47	165			
24	26	Pranav	Roy	37	9	39	46	45	176			
25	27	Rudra	Agarwal	12	35	28	47	38	160			
26	29	Anay	Sharma	4	10	45	37	21	117			
27	20	Ansh	Chatterjee	25	41	5	22	4	107			

low_marks_students

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Clipboard Font Alignment Number												
A1 ID												
	A	B	C	D	E	F	G	H	I	J	K	L
143	171	Vihaan	Chatterjee	3	20	19	34	16	92			
144	172	Krishna	Mehta	4	32	50	43	1	130			
145	173	Sai	Chopra	47	35	46	11	50	189			
146	174	Shaurya	Roy	48	3	9	26	29	115			
147	175	Dhruv	Joshi	22	1	28	46	12	109			
148	176	Aryan	Ghosh	47	17	6	27	29	126			
149	177	Raghav	Patel	29	19	18	12	8	116			
150	178	Aarav	Malhotra	37	11	46	39	27	130			
151	180	Atharv	Iyer	38	38	1	31	25	133			
152	181	Muhamma	Kumar	32	15	35	12	14	108			
153	182	Vivaan	Iyer	15	39	10	34	22	120			
154	183	Aarush	Sharma	20	6	48	35	34	143			
155	184	Aryan	Jain	14	7	5	4	30	60			
156	186	Aditya	Mehta	38	12	24	20	29	123			
157	187	Atharv	Sharma	32	30	5	46	16	129			
158	188	Aarav	Patel	0	47	14	13	17	91			
159	189	Aryan	Chatterjee	37	6	21	16	0	80			
160	190	Anay	Saxena	13	38	7	47	47	152			
161	191	Raghav	Joshi	13	9	14	6	50	92			
162	193	Rudra	Singh	34	6	34	3	39	116			
163	194	Aditya	Rao	7	17	8	15	33	80			
164	195	Aryan	Sen	24	9	45	3	28	109			
165	196	Vivaan	Chopra	3	30	7	31	29	100			
166	197	Reyansh	Patel	0	15	47	44	47	153			
167	198	Om	Agarwal	7	8	4	35	33	87			
168	199	Rudra	Choudhary	37	26	4	20	32	119			
169	200	Aditya	Chatterjee	40	13	9	27	9	98			

low_marks_students